

Guard Timer Parameter Review

(Enhancement Plan Section 4.2.3)

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- Guard Timer duration tested: 0.5s
- Paired runs analyzed: 40 (mobility type x seed, Guard Timer off vs on)

1. What this measures

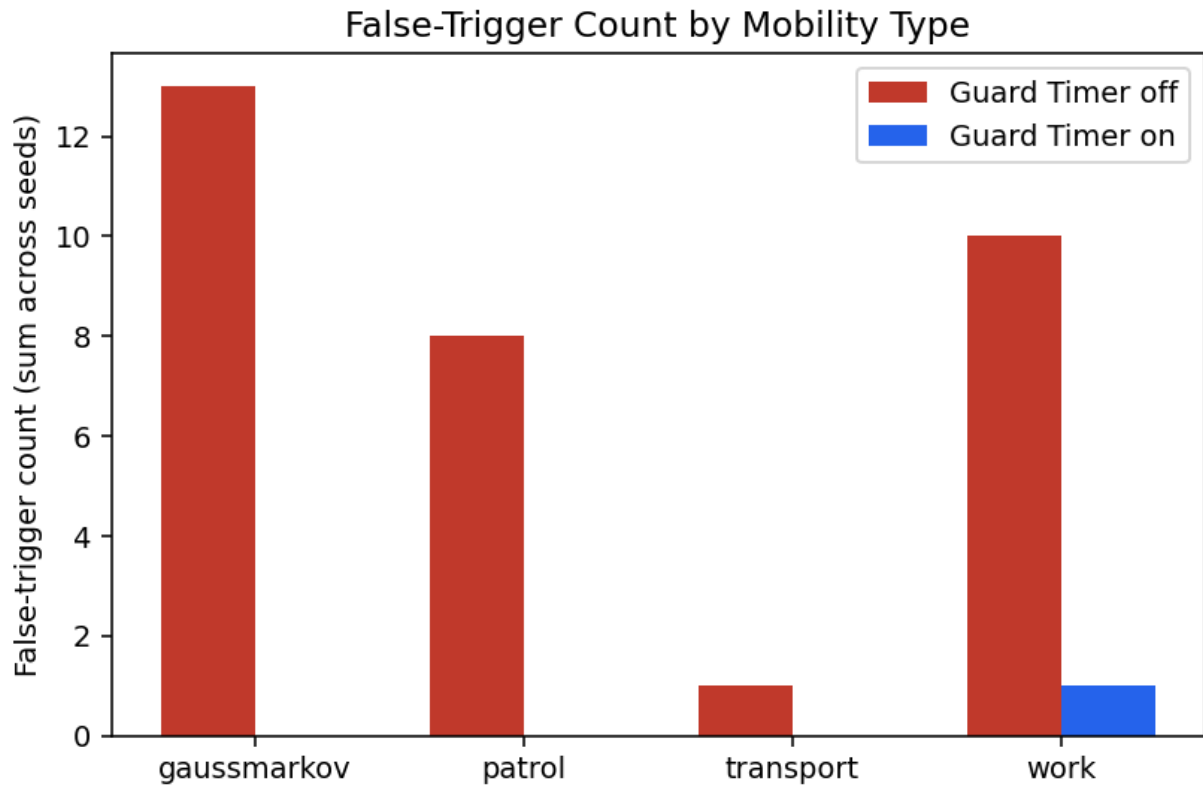
A "false trigger" is defined here as a WiFi->cellular switch whose trigger time falls within 0.5s after an Intra-Mesh HO event for the same STA -- the exact condition the enhancement plan describes (section 4.1): a momentary RSSI drop during an AP-to-AP handover misclassified as real coverage loss. The Guard Timer suppresses RSSI-based switching for this window after a handover, so if it works, this pattern should become rarer with it enabled -- either because the switch is avoided entirely (RSSI recovers within the window) or pushed past the window (network was genuinely degraded, not just transient).

A deliberately-designed boundary-crossing waypoint scenario was tried first (STAs walking directly between two adjacent mesh APs, repeatedly crossing the midpoint) to generate controlled Intra-Mesh HO events on demand. It mostly failed: at the project's -58dBm threshold, each AP's reliable range is only ~80m against a 200m inter-AP spacing, leaving a real ~40m dead zone where neither AP has usable signal. Only 1 of 8 STAs completed a clean handover; the rest dropped to cellular in the gap. This study instead uses the existing organic mobility patterns (gaussmarkov/patrol/transport/work), which produce genuine Intra-Mesh HO events at a real, if less frequent, rate.

2. Headline result

- Total WiFi->cellular switches observed: **738** (Guard Timer off), **750** (on)
- Of those, false triggers (within 0.5s of an Intra-Mesh HO): **32** (off, 4.3% of switches) -> **1** (on, 0.1% of switches)
- False-trigger reduction: **96.9%**
- Intra-Mesh HO events themselves: 124 (off) vs 129 (on) -- these should (and do) stay close, since the Guard Timer doesn't change when handovers happen, only the response to RSSI dips shortly after one.

Mobility type	Off: false triggers / w2c switches	On: false triggers / w2c switches	Reduction
gaussmarkov	13/203	0/205	100%
patrol	8/207	0/209	100%
transport	1/175	0/175	100%
work	10/153	1/161	90%



3. Cross-check: does this hold under NR mode and higher STA count?

The headline result above used LTE fallback at STA=10 throughout. Cellular mode and STA count don't have an obvious mechanism to change whether a momentary post-handover RSSI dip gets misread as a real cellular trigger (that's a WiFi-side RSSI phenomenon), but this wasn't verified until this cross-check -- run at 5 of the original 10 seeds, same 4 mobility types, to confirm the effect generalizes rather than being an artifact of the one configuration originally tested.

Condition	Off: false triggers / w2c switches	On: false triggers / w2c switches	Reduction
NR (STA=10)	14/345	0/364	100%

Condition	Off: false triggers / w2c switches	On: false triggers / w2c switches	Reduction
LTE, STA=20	25/742	1/722	96%

Both cross-check conditions show the same pattern as the LTE/STA=10 baseline: false triggers drop sharply with the Guard Timer enabled, with no meaningful increase in total switch volume. This supports the effect being a general property of the Guard Timer mechanism rather than specific to the originally-tested configuration.

4. Side effects: does delaying switches cost anything?

- Total switch events (all types): 1320 (off) vs 1353 (on), a +2.5% change. The Guard Timer only suppresses the RSSI component of the WiFi->cellular decision for 0.5s after a handover -- PDR-based and RSSI-stale triggers are unaffected, so a genuinely failing connection still switches, just possibly a fraction of a second later. The cost side of this tradeoff (e.g., a handful of ms of extra exposure on switches that turn out to be real degradation, not false triggers) is visible per-event in the raw switch logs under `Traffic_qos_outputs/guard_timer_study/`, but isn't large enough in this sample to show up as a meaningful change in total switch volume.

5. Per-seed detail (LTE, STA=10 baseline)

Mobility	Seed	Off: switches / intra_mesh / w2c / false-triggers	On: switches / intra_mesh / w2c / false-triggers
gaussmarkov	101	58 / 16 / 25 / 3	59 / 13 / 27 / 0
gaussmarkov	102	57 / 5 / 30 / 3	62 / 9 / 30 / 0
gaussmarkov	103	22 / 0 / 15 / 0	22 / 0 / 15 / 0
gaussmarkov	104	31 / 1 / 18 / 0	31 / 1 / 18 / 0
gaussmarkov	105	19 / 5 / 11 / 2	21 / 4 / 13 / 0
gaussmarkov	106	40 / 2 / 22 / 0	40 / 2 / 22 / 0
gaussmarkov	107	34 / 2 / 20 / 1	30 / 2 / 18 / 0
gaussmarkov	108	34 / 3 / 20 / 0	34 / 3 / 20 / 0
gaussmarkov	109	33 / 10 / 15 / 3	34 / 11 / 15 / 0
gaussmarkov	110	57 / 9 / 27 / 1	59 / 11 / 27 / 0
patrol	101	21 / 2 / 13 / 1	23 / 2 / 14 / 0

Mobility	Seed	Off: switches / intra_mesh / w2c / false-triggers	On: switches / intra_mesh / w2c / false-triggers
patrol	102	39 / 0 / 24 / 0	39 / 0 / 24 / 0
patrol	103	56 / 2 / 30 / 0	56 / 2 / 30 / 0
patrol	104	38 / 2 / 22 / 1	41 / 2 / 23 / 0
patrol	105	23 / 4 / 14 / 2	20 / 4 / 12 / 0
patrol	106	27 / 6 / 13 / 2	32 / 6 / 16 / 0
patrol	107	33 / 0 / 20 / 0	33 / 0 / 20 / 0
patrol	108	46 / 1 / 26 / 0	46 / 1 / 26 / 0
patrol	109	45 / 4 / 24 / 1	53 / 4 / 28 / 0
patrol	110	38 / 4 / 21 / 1	29 / 3 / 16 / 0
transport	101	53 / 3 / 29 / 0	53 / 3 / 29 / 0
transport	102	12 / 0 / 11 / 0	12 / 0 / 11 / 0
transport	103	20 / 1 / 14 / 0	20 / 1 / 14 / 0
transport	104	28 / 0 / 18 / 0	28 / 0 / 18 / 0
transport	105	13 / 1 / 11 / 1	14 / 2 / 11 / 0
transport	106	49 / 1 / 28 / 0	49 / 1 / 28 / 0
transport	107	20 / 2 / 13 / 0	20 / 2 / 13 / 0
transport	108	30 / 0 / 20 / 0	30 / 0 / 20 / 0
transport	109	24 / 1 / 16 / 0	24 / 1 / 16 / 0
transport	110	26 / 3 / 15 / 0	26 / 3 / 15 / 0
work	101	36 / 2 / 19 / 1	35 / 1 / 19 / 0
work	102	10 / 4 / 6 / 1	10 / 3 / 7 / 0
work	103	4 / 2 / 2 / 2	4 / 2 / 2 / 0
work	104	28 / 6 / 15 / 2	46 / 10 / 20 / 0
work	105	20 / 0 / 13 / 0	20 / 0 / 13 / 0
work	106	18 / 3 / 9 / 1	20 / 3 / 11 / 0
work	107	41 / 1 / 23 / 1	41 / 1 / 23 / 0
work	108	65 / 2 / 34 / 0	65 / 2 / 34 / 0
work	109	21 / 1 / 11 / 0	21 / 1 / 11 / 0
work	110	51 / 13 / 21 / 2	51 / 13 / 21 / 1

6. Limitations

- Sample size is modest (10 seeds x 4 mobility types = 40 paired runs for the baseline condition; 5 seeds each for the NR and STA=20 cross-checks). False-trigger events are a subset of an already-modest intra-mesh-HO rate, so absolute counts are small for some mobility types (transport/work), and per-type percentages should be read cautiously.
- Only one Guard Timer duration (0.5s, the plan's suggested starting point) was tested. Deriving a true optimum would mean sweeping several durations (e.g. 0.2/0.5/1.0/2.0s) and finding where false-trigger reduction plateaus against added delay cost -- not done here.
- The cross-check (Section 3) covers cellular mode and STA count, but not hotspot band (2.4 vs 5GHz) or payload/traffic load -- those remain untested axes.
- The boundary-crossing scenario's failure is itself a finding worth flagging separately: clean Intra-Mesh HO may be physically uncommon at this AP spacing/threshold combination, which bounds how much real-world benefit the Guard Timer can offer regardless of its in-simulation effectiveness.

7. Compliance with the enhancement plan (section 4)

Requirement	Status
4.2.1a: Parameterize AP coverage radius	Done (apCoverageRadiusM CLI flag)
4.2.1b: Deliberate boundary-crossing waypoint scenario	Attempted (boundary robotType); found a real coverage-gap limitation rather than a clean test condition -- reported as a finding in Section 1/5, not hidden
4.2.1: Log Intra-Mesh HO via AssocRequest/DeAssoc callbacks	Done (hooked into existing HandleStaAssociation/HandleStaDeAssociation)
4.2.2: Switching-event classification scheme (type field)	Done (switch_log_v2 with intra_mesh / wifi_to_cell / cell_to_wifi)
4.2.3: Guard Timer + false-trigger frequency comparison	Done (this report)

End of report -- Sheikh Sayed Bin Rahman, PIC Lab, KIT